How Regulated Systems Produce Responsibility: A Four-Layer Framework for Interaction Design

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Regulated environments — pharmaceutical manufacturing, clinical healthcare, aviation, financial compliance — are typically evaluated as documentation systems: whether records are accurate, procedures are followed, and audit trails are complete. This paper introduces Regulatory UX as a framework that reframes regulated environments as interaction systems and establishes a generalizable mechanism by which system design produces accountability conditions under compliance constraints. The framework identifies four interdependent layers at which design choices determine accountability conditions: an accessibility layer, which shapes what information practitioners can reach; a structural layer, which shapes how document relationships are visible and navigable; a coordination layer, which shapes how work is distributed and observable across actors; and a reasoning layer, which shapes how explanations are generated, compared, and stabilized. Grounded in four prototype-based empirical studies conducted in pharmaceutical GMP environments (each examining one layer, each with $N = 20$ GMP-experienced participants), the framework establishes a consistent mechanism across all layers: system design creates conditions that are invisible to formal quality indicators, that redistribute labor unequally across a workforce stratified by experience, and that attribute the resulting burden to individuals rather than to the structural conditions that produced it. The framework extends existing sociotechnical systems theory and value-sensitive design into compliance-constrained environments, offering an account of how responsibility is produced through system structure rather than assigned after the fact.

Additional Key Words and Phrases: regulatory UX, accountability, compliance-constrained design, sociotechnical systems, GMP, judgment preservation, invisible work, interaction design

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1 Introduction

Regulated environments — pharmaceutical manufacturing, clinical healthcare, aviation, financial compliance — are typically evaluated as documentation systems: whether records are accurate, whether procedures are followed, whether audit trails can be reconstructed. This framing treats regulation as a set of requirements imposed on work, and treats system design as the problem of satisfying those requirements efficiently. It asks whether users perform tasks correctly. It does not ask how system design determines which tasks are available to perform, which interpretations remain open, or which coordination work gets recognized.

I argue that regulated environments should be understood as interaction systems — systems that actively shape how users access information, interpret it, coordinate with others, and ultimately produce accountable decisions. The design of these systems is not neutral with respect to accountability: it determines which explanations remain viable, which

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pathways are visible, and which forms of labor are measured. Responsibility, in this framing, is not only assigned after the fact. It is produced through the structure of the system itself.

I introduce Regulatory UX as a framework for analyzing this production, and derive three design principles that translate it into actionable guidance. Regulatory UX examines how system design structures the conditions under which accountability emerges, treating compliance-constrained interaction as a design problem rather than a compliance problem. The framework identifies four interdependent layers at which system design shapes accountability conditions: an accessibility layer, which determines how users reach information; a structural layer, which determines how document relationships are represented and navigable; a coordination layer, which determines how work is distributed and made visible across actors; and a reasoning layer, which determines how explanations are generated, compared, and stabilized (Figure 1).

The framework is grounded in four empirical studies conducted in pharmaceutical GMP environments, each examining one layer through prototype-based evaluation with $N = 20$ GMP-experienced participants. Full accounts of each study are reported in companion papers currently under review: the accessibility layer study [9], the structural layer study [11], the coordination layer study [8], and the reasoning layer study [10]. This paper synthesizes across the four studies to establish the cross-layer mechanism, introduce the design principles that follow from it, and develop the theoretical argument that regulated environments are interaction systems in which responsibility is produced rather than assigned.

This paper makes three contributions. First, it introduces Regulatory UX as a theoretical framework that reconstitutes regulated environments as interaction systems, shifting the design question from compliance optimization to accountability production. Second, it establishes a four-layer model linking system design to accountability production across access, structure, coordination, and reasoning. Third, it derives three design principles — upstream stabilization, observability over optimization, and judgment preservation — that translate the framework into actionable guidance for designers working in compliance-constrained environments. This positions accountability not as an evaluative outcome, but as a design variable shaped upstream by interaction conditions.

2 Related Work

Sociotechnical Systems and the Design of Work

The premise of the Regulatory UX framework — that system design shapes the conditions under which accountability emerges — draws on a long tradition of sociotechnical systems research. Baxter and Sommerville [2] establish that the design of work systems cannot be separated from the social conditions in which they operate: technical design choices produce organizational outcomes, and organizational conditions shape how technical systems are used. Their framework for sociotechnical systems engineering emphasizes that system design is not simply a technical process but a process that co-determines how work is organized, how labor is distributed, and who bears the cost of system failures. Regulatory UX extends this argument into compliance-constrained environments specifically — environments in which the sociotechnical stakes of design are particularly high because accountability is formally required and legally consequential.

Values in Design

Friedman et al.'s (2002) Value Sensitive Design provides a complementary theoretical foundation. VSD establishes that technology design is never value-neutral: design choices embed assumptions about who matters, what counts as

success, and how competing interests are balanced. Regulatory UX applies this insight to accountability specifically: in regulated environments, design choices embed assumptions about how responsibility is distributed, which forms of labor are recognized, and whose epistemic access to information is treated as a design concern.

Documents as Organizational Action

Harper's (1998) ethnographic account of documentation practices at the IMF establishes the foundational premise of the framework's structural layer: documents are not neutral containers of information but active participants in organizational action. Their accessibility, their linkage to other documents, and the navigational demands they impose on users are constitutive of organizational practice. When document relationships are visible and traversable, practitioners can reconstruct accountable pathways; when they are not, that reconstruction work falls to individuals.

Awareness and the Visibility of Work

Dourish and Bellotti's (1992) account of awareness in collaborative work systems establishes the theoretical basis for the framework's coordination layer. They show that awareness — understanding of the activities of others that provides context for one's own action — is critical to successful coordination and is poorly served by mechanisms that require explicit information-sharing. The framework's coordination layer extends this insight: it treats the invisibility of review friction not as a communication failure but as a design failure — a condition produced by systems that do not provide the awareness infrastructure that coordination requires.

Invisible Work and the Attribution of Labor

Star and Strauss [12] establish that work performed in the gaps of formal systems tends to become invisible — not captured by indicators, not attributed to the conditions that produced it, and not recognized as labor by the organizations that depend on it. This insight runs through all four layers of the Regulatory UX framework. At the accessibility layer, retrieval burden is invisible upstream of formal errors. At the structural layer, traceability reconstruction is invisible in quality metrics. At the coordination layer, review friction is invisible in closure records. At the reasoning layer, comparison collapse is invisible in final investigation documents. In each case, system design creates conditions that redistribute labor, and the redistribution goes unmeasured because the labor is performed in the gaps between formal indicators.

Situated Action and the Conditions of Accountable Practice

Suchman's (1987) account of situated action provides the theoretical spine of the framework. Suchman establishes that action is not plan-driven but shaped by the immediate conditions of its execution — what is visible, accessible, and interpretable at the moment a decision must be made. Regulatory UX applies this principle systematically across four layers: at each layer, the conditions of action shape the outcomes of action, and those conditions are themselves shaped by system design.

Human-AI Interaction and the Preservation of Judgment

The framework's reasoning layer draws on Amershi et al.'s (2019) guidelines for human-AI interaction, which establish that AI systems should support human control, make their limitations transparent, and avoid pre-empting human judgment. In regulated environments, this principle is not optional — accountability requirements mandate that quality decisions be made by qualified human reviewers, and any system that automates those decisions introduces a compliance

risk. The framework's judgment preservation principle extends this guideline into a design constraint: in compliance-constrained environments, the goal of AI support is not to accelerate decisions but to preserve the conditions under which genuine human judgment is possible.

Epistemic Access and the Distribution of Knowledge

Fricker's (2007) account of epistemic injustice provides theoretical grounding for the framework's unifying theme. Fricker establishes that epistemic access — the ability to know, to be heard as a knower, and to participate in the production of knowledge — is not equally distributed, and that structural conditions determine who can contribute to and benefit from shared knowledge practices. Applied to regulated work systems, this framework implies that design choices which restrict information access, obscure document relationships, or suppress comparison are not merely usability failures: they are conditions that determine whose knowledge counts and who bears the cost of structural invisibility.

AI-Assisted Decision Making and Judgment-Preserving Design

Within HCI, work on AI-assisted decision making has established that interface structure actively shapes which information becomes salient to practitioners navigating complex regulated environments. Wang and Yin [14] demonstrate that explanation interfaces alter how decision makers engage with outputs depending on domain expertise — findings that establish regulated interfaces as active shapers of accountability conditions rather than neutral conduits. Green and Chen [6] show that algorithm-in-the-loop settings systematically constrain the conditions under which independent judgment is possible, grounding the framework's judgment preservation principle: when interface structure determines what practitioners can see and act on, design choices that preserve interpretive authority are not merely usability preferences but structural requirements of compliance-constrained environments.

Prior work has not synthesized these theoretical resources into a unified framework for the design of compliance-constrained interaction environments. The Regulatory UX framework developed in this paper makes that synthesis explicit, grounding it in four empirical studies that demonstrate the mechanism at each layer.

3 The Regulatory UX Framework

Overview

Regulatory UX is a framework for analyzing how system design shapes accountability conditions in compliance-constrained environments. It identifies four interdependent layers at which design choices determine the conditions under which accountable action is possible: an accessibility layer governing information reach, a structural layer governing document relationship visibility, a coordination layer governing the observability of collaborative work, and a reasoning layer governing how explanations are generated and stabilized (Figure 1). Each layer addresses a distinct aspect of the relationship between system structure and accountability outcome; together, they describe a mechanism through which responsibility is produced — not assigned after the fact, but structured in advance by the design of the systems through which work is done.

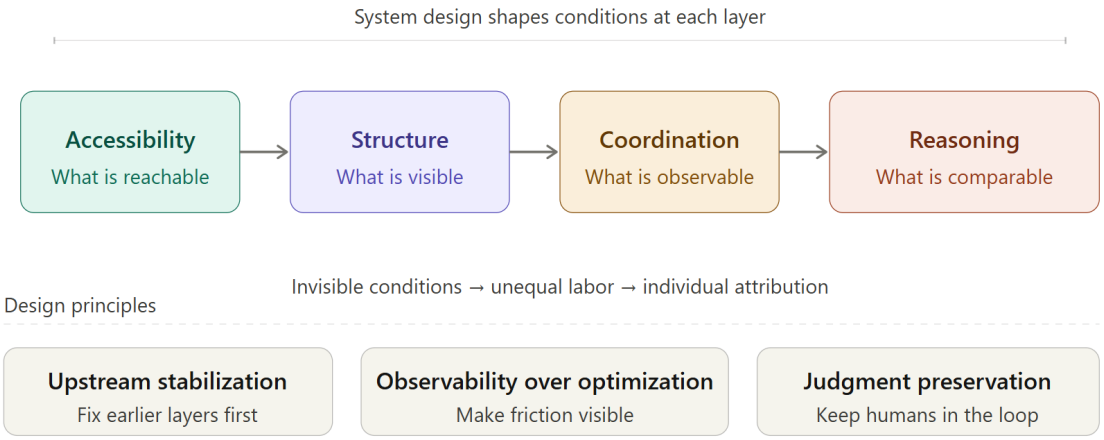


Fig. 1. The Regulatory UX framework: four interdependent layers at which system design shapes accountability conditions in compliance-constrained environments. Each layer addresses a distinct aspect of the relationship between system structure and accountability outcome. Arrows indicate the direction of causal dependency: conditions at earlier layers shape the conditions available at later ones.

The framework is grounded in a consistent theoretical commitment: following Suchman [13], it treats action as situated — shaped by the immediate conditions of its execution rather than by formal plans or procedures alone. In regulated environments, those conditions are substantially determined by system design. The framework makes this determination explicit and tractable, identifying four layers at which design choices shape the conditions that shape the action that shapes accountability.

Layer 1: Accessibility

The accessibility layer concerns how practitioners reach information — the conditions under which documents, procedures, and data are locatable, retrievable, and ready-to-hand at the moment of action.

In regulated environments, accessibility is structurally constrained. Documents may be stored as scanned files that block keyword search; archive organization may reflect historical naming conventions rather than task-relevant categories; validated source files may not be modifiable without triggering change control processes. These constraints are not incidental — they are constitutive features of regulated document environments that shape what a practitioner can access, and therefore what they can know, at any given moment of practice.

The accessibility layer connects to accountability conditions through a specific mechanism: when access is constrained, interpretation begins from incomplete information. A practitioner who cannot locate the most applicable procedure makes decisions under conditions that the system has structured. The resulting interpretive variability is upstream of any formal error — it precedes the action that will later be evaluated for compliance.

Design implication: accessibility interventions should operate externally, over metadata or retrieval infrastructure, without modifying validated source documents. The design challenge is to reduce navigational burden without introducing compliance risks that flow from modifications to validated content. In practice, this means keyword-based retrieval

layers that decouple search from archive naming conventions, result cues that support document discrimination before opening, and empty-result transparency that distinguishes wording mismatches from genuine archive gaps.

Layer 2: Structure

The structural layer concerns how document relationships are represented — the conditions under which connections between documents are visible, navigable, and reconstructable.

In regulated environments, documents form interdependent networks: SOPs reference method protocols; deviation records link to CAPA documents; validation artifacts connect to specification requirements. These relationships are typically recorded as textual references rather than navigable links, and their maintenance is a secondary concern relative to document content. When references are incomplete, outdated, or ambiguous, practitioners must reconstruct pathways manually — bridging the gaps between documents through inference and memory rather than through structural navigation.

The structural layer connects to accountability conditions through the distribution of invisible labor. When linkage structure is incomplete, the work of reconstructing accountable pathways is performed by individuals, not by systems. This work is distributed unequally across a workforce stratified by experience: practitioners who have internalized a map of the document network bear less of this burden; those who have not bear more. In the structural layer study [11], less experienced participants averaged more than three times as many navigation loopbacks as experienced practitioners — not because they reasoned poorly, but because they lacked the internalized map that the document system does not provide.

Design implication: structural interventions should make linkage relationships navigable without modifying validated source files. Graph-based visibility layers that surface document relationships, flag broken links, and support traceability reconstruction address the structural layer without introducing document modification risks.

Layer 3: Coordination

The coordination layer concerns how work is distributed and made visible across actors — the conditions under which the labor of collaborative processes is observable, attributable, and discussable.

In regulated environments, review workflows involve multiple actors cycling through iterative rounds of comment, revision, and approval. The coordination work that produces these cycles is typically invisible: it is not captured by workflow indicators, not measured by quality systems, and not attributed to the structural conditions that produce it. When delays occur, they are experienced asymmetrically — felt by document submitters, invisible to quality metrics, and attributed to individuals rather than to coordination conditions.

The coordination layer connects to accountability conditions through visibility. When the conditions of review are not visible — when revision burden, comment distribution, and bottleneck formation cannot be seen as patterns — accountability for delay is assigned without access to the coordination conditions that produced it. In the coordination layer study [8], all 20 participants anticipated reduced accountability asymmetry from friction visibility, describing the dashboard as infrastructure for making coordination asymmetry discussable rather than invisible.

Design implication: coordination interventions should surface workflow patterns as interpretable signals without modifying validated documentation systems or automating quality decisions. The design challenge is to make friction visible while preserving the human judgment that is required for quality decision-making in compliance-constrained environments.

Layer 4: Reasoning

The reasoning layer concerns how explanations are generated, compared, and stabilized — the conditions under which causal accounts remain open to revision or close prematurely around a single explanation.

In regulated environments, investigation workflows impose closure pressure: temporal constraints, documentation requirements, and accountability expectations that define when an investigation must be complete. Under this pressure, comparison between alternative explanations collapses before reasoning has fully developed. The resulting causal account reflects the conditions of its production: an explanation that satisfies documentation requirements under closure pressure, not necessarily one that has been systematically compared to alternatives. In the reasoning layer study [10], all 20 participants reported selecting explanations oriented toward documentation acceptability rather than investigative completeness — choosing contained, person-centered accounts that would satisfy audit review without opening systemic lines of inquiry.

Design implication: reasoning interventions should preserve comparison between alternatives by structuring investigation workflows to require explicit comparison before closure. AI support in this layer should function as a comparison enforcer rather than an explanation source — detecting convergence signals and reopening the comparison space rather than suggesting conclusions.

The Mechanism: From Design to Accountability

Across all four layers, a consistent mechanism operates. System design shapes the conditions of action at each layer; the conditions at each layer shape the conditions at the next; and the cumulative effect of design choices across all four layers determines how responsibility is produced and distributed.

At the accessibility layer, design determines what information is reachable. At the structural layer, design determines what relationships are visible. At the coordination layer, design determines what labor is observable. At the reasoning layer, design determines what explanations remain comparable. In each case, design choices that leave conditions invisible, inaccessible, or unsupported redistribute labor and accountability to individuals — while the conditions that produced the redistribution remain invisible in formal records.

This mechanism is the basis for the framework's central claim: in regulated environments, responsibility is not only assigned at the end of a process. It is produced, layer by layer, through the structure of the systems in which that process unfolds.

4 Empirical Program

The Regulatory UX framework is grounded in four prototype-based studies conducted in pharmaceutical GMP environments. Each study examines one layer of the framework, using a combination of synthetic document corpora, behavioral observation, and pilot evaluation with $N = 20$ GMP-experienced participants. The studies are reported in full in companion papers [8–11]; this section summarizes the key findings at each layer and their contribution to the framework.

Layer 1: Accessibility — SOP Retrieval and Interpretive Burden [9]

The accessibility study examined how retrieval constraints shape interpretive practice in GMP quality control environments. A synthetic corpus of 500 GMP SOPs was constructed and indexed, and an external keyword-searchable retrieval layer was developed and evaluated with $N = 20$ frequent SOP users using a think-aloud protocol.

The central finding was not that a retrieval tool reduces search time — though it does, from approximately six minutes to approximately one minute under exploratory conditions — but that access conditions shape the behavioral repertoire practitioners allow themselves. When SOP retrieval is structurally difficult, practitioners act within a personally known subset of the procedural archive, consulting procedures they have used before rather than those that might be most applicable. 18 of 20 participants rated the prototype as intuitive, and all 20 indicated willingness to reuse the system in live archive conditions.

The study also surfaced a category of invisible onboarding labor — informal SOP guidance provided by senior staff to resolve nomenclature ambiguities — that the quality system does not measure. This labor sits upstream of any formal error record and is not attributed to the archive structure that produces it.

Layer 2: Structure — Document Traceability and Navigation Burden [11]

The structural study examined two related questions: whether making document linkage structure visible reduces traceability burden, and whether behavioral navigation tracking can surface document-structural friction as an observable signal. A synthetic GMP document corpus was analyzed as a directed graph using NetworkX — identifying 31 missing CAPA links, 103 broken downstream links, and 21 lifecycle mismatch risks — and a behavioral navigation tracking prototype was evaluated with $N = 20$ GMP-experienced participants during a standardized out-of-specification investigation scenario.

The traceability visibility prototype was associated with a 30–60% reduction in path reconstruction time; all 20 participants successfully completed target pathway reconstruction, rated ease of use at 5.0/5, and expressed adoption intent. The navigation tracking study found that loopback behavior was observed across all 20 participant sessions, with a mean of approximately 5 loopbacks per session. Experienced practitioners averaged 2–3 loopbacks, while less experienced participants averaged 7–10; all participants attributed navigation difficulty to document structure rather than interface design.

Layer 3: Coordination — Review Friction and Workflow Visibility [8]

The coordination study examined whether workflow friction in deviation and CAPA review processes can be externalized as interpretable signal. A synthetic dataset of 100 review records was constructed, and a six-tab workflow dashboard combined with a TF-IDF pattern detection layer was evaluated with $N = 20$ QA/QC-experienced participants. All 20 participants rated both components 5.0/5 on usability and expressed adoption intent.

The central finding was that all 20 participants anticipated reduced accountability asymmetry from friction visibility. Participants described the dashboard as infrastructure for making coordination asymmetry discussable — not by resolving it automatically, but by making it visible as a pattern. The study also identified an organizational adoption barrier: participants questioned whether regulated organizations would deploy a system that makes review reasoning more transparent, given that transparency surfaces conditions that current practice leaves invisible.

Layer 4: Reasoning — Premature Convergence and Comparison Preservation [10]

The reasoning study examined how closure pressure shapes causal reasoning in GMP deviation investigations, and whether structured scaffolding can preserve comparison between alternatives. A timed reflection protocol ($N = 20$, three timepoints) established that comparison is time-sensitive: 15 of 20 participants produced single-cause, person-centered explanations at the initial timepoint, with confidence ratings averaging 1 out of 5 across all participants regardless of

experience level — indicating that early convergence reflects institutional pressure rather than epistemic certainty. 18 of 20 participants broadened their causal accounts within ten minutes of unstructured reflection.

An AI-supported reasoning scaffold organized investigation into five stages enforcing explicit comparison before closure. 18 of 20 participants produced broader causal accounts than in the unassisted condition; 15 of 20 reconsidered their initial attribution, with 12 of those 15 shifting from person-centered to systemic or procedural explanations. All 20 participants expressed adoption intent while simultaneously identifying the audit trail implications of transparent investigation reasoning as a structural barrier to deployment.

Cross-Layer Synthesis

Across all four studies, a consistent structural pattern emerges. In each layer, system design creates conditions that are invisible to formal quality indicators. In each case, the invisible condition produces labor that is absorbed by individuals, distributed unequally across experience levels, and attributed to individual failure rather than structural design. And in each case, a prototype intervention demonstrates that making the invisible condition visible reduces the labor burden — without modifying validated systems, automating quality decisions, or displacing human judgment.

This cross-layer pattern is the empirical basis for the framework's central claim: regulated environments produce accountability outcomes through the structure of their systems, not only through the decisions of their practitioners. What the pattern adds to prior sociotechnical accounts is specificity: it is not that system design influences accountability in general, but that a single mechanism — structural invisibility producing redistributed, unattributed labor — operates consistently across four analytically distinct layers of regulated work, each addressable through design intervention without altering the compliance constraints that govern the environment.

5 Design Implications

The Regulatory UX framework suggests three design principles that translate the four-layer analysis into actionable guidance for designers working in compliance-constrained environments.

Principle 1: Upstream Stabilization

The first principle follows from the layered structure of the framework: improving accountability outcomes requires stabilizing conditions at earlier layers before intervening at later ones. A practitioner who cannot locate the relevant SOP (accessibility layer failure) cannot engage in reliable traceability reconstruction (structural layer); a practitioner navigating an opaque document network cannot participate effectively in review coordination (coordination layer); a practitioner operating under unresolved retrieval and traceability burdens cannot sustain meaningful comparison in investigation reasoning (reasoning layer).

In practice, this means that retrieval system design is not a secondary concern to be addressed after investigation workflow support has been implemented — it is a precondition for that support to function. The principle has a practical implication that conventional usability approaches do not: it is not sufficient to improve the interface at the point of decision-making if the conditions upstream of that decision have already constrained what is available to be decided.

Principle 2: Observability over Optimization

The second principle reorients the conventional UX goal of efficiency optimization. In regulated environments, making work faster is not always an improvement: faster retrieval that bypasses validation constraints introduces compliance risk; faster investigation closure that meets documentation deadlines may produce premature convergence. The goal of

design in compliance-constrained environments is not to minimize the time spent on work but to make the work that is being done visible — to surface the conditions, labor, and patterns that formal indicators leave invisible.

This principle distinguishes Regulatory UX from conventional usability engineering in a specific way. Usability engineering optimizes for task performance: it measures completion time, error rates, and satisfaction, and designs to improve them. Regulatory UX optimizes for observability: it asks what is currently invisible in the system — which labor is unattributed, which conditions are unmeasured, which patterns are distributed across actors without a shared representation — and designs to make those things visible. The appropriate evaluation metrics are not task completion time or error rate, but: what was made visible that was previously invisible, how the distribution of labor changed, and whether accountability conditions shifted.

Principle 3: Judgment Preservation

The third principle follows from the compliance requirements of regulated environments: any system intervention that automates quality decisions, pre-empts human accountability, or produces outputs that function as quality conclusions introduces a regulatory risk. In pharmaceutical manufacturing, quality decisions must be made and documented by qualified human reviewers; the accountability chain requires that a human decision-maker can be identified for every quality conclusion that enters the formal record.

This principle constrains what AI can do in regulated environments more stringently than Amershi et al.'s (2019) general guidelines for human-AI interaction. It is not merely that AI should support rather than replace human judgment; it is that in compliance-constrained environments, any design that blurs the boundary between AI output and human judgment creates a traceability problem for the accountability record itself.

Judgment preservation has different implications at each layer. At the accessibility layer, it means retrieval tools surface documents without recommending them. At the structural layer, it means visibility tools surface linkage patterns without interpreting them. At the coordination layer, it means workflow tools surface friction patterns without assigning responsibility for them. At the reasoning layer, it means AI detects convergence signals and reopens comparison without suggesting which alternative is correct.

The Tension Between Principles and Organizational Incentives

A limitation of these principles, revealed by the empirical studies, is that they can come into conflict with organizational incentive structures. Upstream stabilization requires preventive design investment that is difficult to justify without evidence of the problems it prevents. Observability over optimization may surface conditions that organizations prefer to leave invisible. Judgment preservation may be resisted in favor of AI systems that accelerate closure, because closure is what regulated organizations are formally rewarded for.

These tensions are not design failures; they are features of the organizational environments in which Regulatory UX principles are meant to operate. The framework identifies where the tensions arise; resolving them is an organizational and institutional challenge as much as a design one.

6 Discussion

Regulatory UX as a Reframing

The dominant framing treats regulation as a constraint on design — a set of requirements that limit what designers can do and that usability engineering must work around. Regulatory UX inverts this framing: it treats regulation as the context

in which accountability is produced, and treats design as a participant in that production. The question is not how to design systems that satisfy regulatory requirements; it is how to design systems that shape accountability conditions in ways that are fair, observable, and supportive of genuine human judgment. This inversion is the framework's theoretical contribution: it is not a synthesis of existing sociotechnical accounts but a respecification of the design problem — from compliance optimization to accountability production — that those accounts, individually, do not reach.

Epistemic Access as a Unifying Theme

Across all four layers, the empirical studies converge on a finding that the framework makes explicit: system design in regulated environments determines epistemic access — who can see what information, navigate which pathways, observe which patterns, and sustain which comparisons. The practitioners who bear the greatest burden at each layer are consistently those with the least accumulated implicit knowledge of the system. This asymmetry is not a training gap. It is a design gap. The implicit knowledge that experienced practitioners carry is knowledge that the system should provide structurally — the navigational support, the linkage visibility, the coordination awareness, the comparison scaffolding — but currently externalizes into human memory and informal practice.

The experience asymmetry is most sharply evidenced in the structural layer, where less experienced participants averaged more than three times as many loopbacks as experienced practitioners — not because they reasoned poorly, but because they lacked the internalized map of the document system. The same asymmetry appears at the reasoning layer, where confidence ratings of 1 out of 5 at the initial timepoint reveal that the institutional conditions producing premature convergence operate uniformly across experience levels. Fricker's (2007) account of epistemic injustice is directly applicable here: the structural conditions that restrict epistemic access are not distributed by competence but by accumulated implicit knowledge that the system itself does not provide.

The Limits of Interface Design

The organizational adoption barriers identified across the empirical studies point to a limitation of the framework that interface design alone cannot address. At the coordination layer, participants questioned whether organizations would deploy tools that make review friction visible. At the reasoning layer, all 20 participants identified the audit trail implications of transparent investigation reasoning as a structural barrier to deployment — not because the scaffold was poorly designed, but because transparency surfaces conditions that current practice leaves invisible. In both cases, the concern was not that the tools were poorly designed but that transparency — the core value the tools instantiate — may not be organizationally incentivized.

The framework can identify where accountability conditions are shaped by invisible design choices, and it can demonstrate that making those conditions visible is technically feasible. What it cannot do is change the organizational incentive structures that make invisibility rational. Addressing those structures requires engagement with quality culture, audit standards, and institutional accountability — work that is organizational and regulatory as much as it is design work.

Generalizability Beyond Pharmaceutical Manufacturing

The framework was developed in pharmaceutical GMP environments, but the mechanism it identifies is not specific to that domain. Any regulated environment in which accountability is formally required and system design is consequential for how that accountability is produced is a candidate for Regulatory UX analysis: clinical healthcare, aviation

maintenance, financial compliance, legal practice, and governance systems all share the structural features that make the framework applicable.

Future Directions

The framework opens several directions for future work. First, the organizational adoption problem identified at the coordination and reasoning layers warrants systematic investigation: under what conditions do regulated organizations value transparency over opacity, and what organizational or institutional changes make Regulatory UX interventions sustainable? Second, the cross-layer dependencies identified in the upstream stabilization principle need empirical examination: to what extent do accessibility and structural interventions actually improve coordination and reasoning outcomes? A specific testable hypothesis follows from the framework: practitioners with reduced loopback behavior (structural layer) should also show reduced premature convergence rates (reasoning layer), because the navigational burden absorbed at the structural layer is no longer available to constrain reasoning. Third, the framework should be applied and tested in domains beyond pharmaceutical manufacturing.

7 Conclusion

This paper introduced Regulatory UX as a framework for explaining how regulated systems produce responsibility through interaction conditions. Against the dominant framing of regulated environments as documentation systems, the framework argues that regulated environments are interaction systems, and that design choices at four interdependent layers determine how responsibility is produced before it is ever formally assigned.

The four-layer model — accessibility, structure, coordination, and reasoning — identifies a consistent mechanism across all layers: system design creates conditions that are invisible to formal quality indicators, redistribute labor unequally across a workforce stratified by experience, and attribute the resulting burden to individuals rather than to the structural conditions that produced it. Four prototype-based empirical studies ground this mechanism at each layer, establishing that the invisible conditions are real, that they produce measurable labor and friction, and that making them visible is technically feasible without modifying validated systems or displacing human judgment.

Three design principles follow. Upstream stabilization holds that accountability conditions must be addressed at earlier layers before interventions at later ones can be effective. Observability over optimization holds that the design goal in compliance-constrained environments is not efficiency but visibility. Judgment preservation holds that in regulated environments, the role of computational support is to improve the conditions under which human judgment operates, not to substitute for it.

The framework's limits are also clear. Interface design cannot change the organizational incentive structures that make invisibility rational. Addressing those structures is an organizational and institutional challenge as much as a design one. Regulatory UX identifies where those challenges arise and provides the analytical vocabulary for engaging them; translating that engagement into organizational change requires work that extends beyond the scope of interaction design.

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